Environmental Economics: An Introduction Barry Field, 2024 Release (Chapter 8)

Answers to Discussion Questions

1. If one focuses only on the directly impacted industry, that is apparently what it means. But, as we have tried to stress throughout the book, enforcement costs have to be considered. If enforcement costs were to remain unchanged, the implication is true, but with a different regulation they may change, in which case the total costs of the changed standard will be different.

2. The costs would be measured in terms of the development value the land would have that is foregone. If one were to make the assumption that no development will take place as a result of the restriction, then the full value would be a cost. If one assumed that development is diverted to another location, the lost value of development would be the difference in developed values between the protected location and the spot to which development is diverted. Note: The protecting of the wetland would be a social benefit.

3. A project-type question. Essentially, the analyses should reveal that what may be relatively easy for one person with a low opportunity cost of the activity can be harder to implement on a broader and/or longer scale with greater opportunity costs.

4. The problem here is assuming that the same quantity of gas will be sold even though the price is 10¢ a gallon higher. But a price rise of this amount would lead, other things being equal, to a reduction in quantity demanded, the amount of which depends on the price elasticity of demand. Of course, other things are never equal in the real world, for example, population growth may occur, which would tend to offset part or all of the price increase. Refer to Figure 3.3, which shows the inverse relationship of price and quantity.

5. This makes cost estimation for a single industry much more complicated. If all firms were alike, it would be possible simply to take a look at the costs of a representative firm and then multiply the results by the number of firms. If there are relatively few types of firms in the industry (e.g., small vs. large), one could do the same thing as a weighted average of the number of firms in each category. If the firms are truly all very heterogeneous, one will simply have to either study them all or make an estimate as best as one can.